TUSHAR BHARADWAJ

📍 Bangalore, India | 📞 +91 98828 78776 | ✉️ ttushar684@gmail.com

🔗 LinkedIn | GitHub | Website

PROFESSIONAL SUMMARY

Remote Sensing and GIS Lead with a total of 10 years of professional experience, including 6.5 years specializing in GIS and Remote Sensing. Currently working as GIS Lead at Thryve Earth, driving geospatial analytics, carbon intelligence tools, and data-driven decision systems. Experienced in geospatial data analysis, satellite imagery processing, spatial databases, and analytical modeling using Python and SQL. Strong track record of translating complex spatial datasets into actionable insights through advanced geospatial analytics, visualization, and scalable data workflows. Extensive experience working with satellite data (optical, SAR, LiDAR) and building analytics pipelines for climate and carbon monitoring systems.

CORE SKILLS

Machine Learning & AI: Applied ML, Feature Engineering, Model Validation, Geospatial ML
Data Analytics: Python, SQL, Geospatial Data Analysis, Statistical Analysis, Data Interpretation
Geospatial & Remote Sensing: Satellite imagery, SAR, LiDAR, Spatial Analysis, GIS Mapping
Visualization & Tools: Google Earth Engine, QGIS, ArcGIS, Geospatial Dashboards, Data Visualization

WORK EXPERIENCE

Thryve Earth - Remote Sensing and GIS Lead (Contract)

Mar 2024 – Present

- Own ML strategy, planning, and productization of carbon and geospatial intelligence tools designed for external commercialization.
- Automation of Carbon modeling for various project types.
- Architect scalable ML systems for biomass estimation, land-use change, and carbon accounting aligned with VCS & Gold Standard.
- Develop and deploy ML/CV models for forest cover change, degradation detection, and ARR/REDD+ prospecting.
- Lead automation of high-throughput satellite data pipelines using Python, GEE, and GCP.
- Work closely with product, engineering, and carbon teams to embed ML into core platforms.
- Translate ambiguous business and carbon methodology requirements into production-ready ML solutions.

Plant-for-the-Planet - Remote Sensing Data Scientist (Part-time)

Apr 2022 – Mar 2024

- End-to-end ownership of ML and geospatial intelligence across the organization.
- Led development of the Tropical Forest Forever Facility (TFFF) Watch Tool from concept to deployment.
- Built ML-driven biomass monitoring and time-series change detection pipelines.
- Designed EUDR compliance systems integrating land-use, deforestation, and commodity datasets.
- Developed automated tools for fire, flood, and afforestation monitoring.

IIT Roorkee - Research Fellow (ML & Computer Vision)

Apr 2022 – Mar 2024

- Built ML and CV pipelines for urban forestry using multispectral and LiDAR data.
- Developed tree canopy extraction and biomass estimation models.
- Led model experimentation, validation, and benchmarking.
- Integrated ML workflows with cloud-based remote sensing ETL pipelines.
- Created interactive dashboards using GEE and JavaScript.

Govt. College of Technology and Engineering - Assistant Professor (GIS & RS)

September 2019 – February 2022

- Taught ML-driven geospatial analytics, Python-based data analysis, and GIS.
- Supervised applied ML and remote sensing projects.
- Mentored students on real-world data science problem-solving.

Amar Found – Senior Geotechnical Engineer

May 2019 – September 2019

- Executed projects involving diaphragm walls, soil anchors, and topographic surveying.

Lovely Professional University – Assistant Professor

August 2017 – April 2019

- Delivered courses in Civil Engineering and supervised research on soil stabilization using agri-waste.
- Contributed to syllabus development and curriculum modernization.

EDUCATION

- GIS and Remote Sensing Courses, IIT Roorkee (9.5/10)
- Post Graduate Diploma in Data Science, IIIT Bangalore (3.12/4.0)

- M.Tech – Geotechnical Engineering, NIT Hamirpur (7.6/10)
- B.Tech – Civil Engineering, Rajasthan Technical University (64.8%)

TECHNOLOGY STACK

ML & AI: Python, Scikit-learn, TensorFlow, OpenCV, CNNs, RF, SVM, Time-Series
Cloud & MLOps: GCP, BigQuery, Cloud Storage, ML Pipelines
Geospatial: Google Earth Engine, GDAL, Rasterio, GeoJSON, QGIS, ArcGIS, PostgreSQL
Carbon & Climate: Biomass modeling, MRV workflows, REDD+, ARR, EUDR

PROJECT HIGHLIGHTS

Carbon Project Prospecting Platform – ML-based ARR/REDD+ suitability analysis
Global Biomass Estimation Model – GEDI + SAR + optical ML pipeline
EUDR Compliance Tool (Tracer) – Production geospatial ML web application
Urban Tree Species Mapping – CV-based classification system

PUBLICATIONS

Bharadwaj, T., Kumar, A., & Khare, S. (2023). Smart agriculture platforms: Case studies. In *Technological Prospects and Social Applications of Society 5.0* (pp. 169–176). CRC Press. https://doi.org/10.1201/9781003324720-15

Khare, S., Bharadwaj, T., Wood, S. L. R., Theron, J., Filotas, E., & Gonzalez, A. (2023). "Machine learning and satellite-based multi-scale segregation of coniferous and deciduous trees: a case study for Montreal city, Canada." GEO BON Global Conference: Monitoring Biodiversity for Action.

LANGUAGES

English (Fluent), Hindi (Fluent)

PERSONAL DETAILS

Date of Birth: 19 Sept 1994
Nationality: Indian